

Living with Smell Dysfunction: A Multi-sensory VR Experience

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ABSTRACT

"Living with Smell Dysfunction" is a multi-sensory short film that introduces scents in Virtual Reality (VR) Experience. Through this first-person immersive film in VR, the participant will face daily adventure, confusion and danger as a patient with smell dysfunction. Olfactory disorders simulated in this film are anosmia (absence of smell), hyposmia (diminished sensitivity of smell), and dysosmia (distortion of normal smell). [Schiffman 2007] Olfactory dysfunction and disability are tend to be overlooked and invisible to vast population. This novel immersive experience could bring more discussion and attention to the treatment and daily lives of smell dysfunctional patients.

CCS CONCEPTS

• Human-centered computing;

KEYWORDS

Virtual Reality, Smell dysfunction, Covid-19, Scent, Multi-sensory

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1 INTRODUCTION

Before COVID-19 pandemic, 1 in 8 Americans over age 40 had measurable smell dysfunction. [Hoffman et al. 2016] As the COVID-19 pandemic spreads, more and more cases of lasting olfactory dysfunction are reported, creating an unprecedented worldwide demand for treatment. Sudden smell loss is a specific early symptom of COVID-19, with an estimated prevalence of 40% to 75%. [Ohla et al. 2021] People with smell loss are twice as likely to experience a smell-related hazardous event. [Pence et al. 2014] People who have lost their sense of smell also report high rates of depression, anxiety, isolation, and relationship difficulties. [GF and JY 2016] Smell disorder symptoms are emerging as new common long-term conditions. [Ohla et al. 2021]

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"Living with smell dysfunction" is an immersive multi-sensory VR experience exploring the everyday struggles of those with smell disorders. Using novel techniques of movement detection, real-time scent mapping, 3D animations, 360° videos, we put the audience into the shoes of people with smell disorders.

2 DESIGN AND IMPLEMENTATION

In this section we introduce our hardware and software implementation from two parts, including electrical structure of scent releasing device and user interaction in VR.

2.1 Electrical Structure

We have a customized atomizer device to support the scent releasing process during the film. This atomizer unit consists of a mini PCB driver, piezoelectric ceramic disc, metal diaphragm, and a silicone cover. There are several iterations of the unit, some are for wearable devices, the others are for stationary installation. (Fig. 1a) The wearable device in display is 3.15 x 3.15 x 0.78 inches in size. (Fig. 1b) The material is 405nm SLA UV-curing clear resin. Currently, our unit supports the control of up to 8 different scents. To precisely control the scent-delivering process with Unity in real-time, we are adapting Bluetooth communication (BLE) in our latest prototype for iOS platform. For this VR experience, the scent releasing module requires a physical wire for Arduino and Unity to communicate via the serial port on PC. The device could be held by hand or mounted on the VR headset as an add-on accessory.

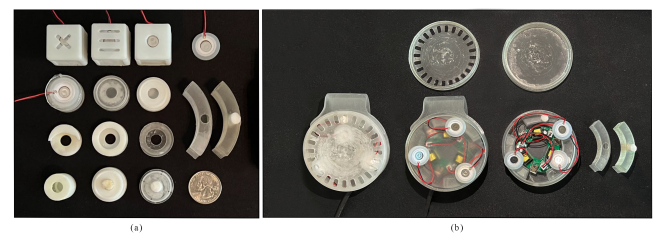


Figure 1: Wearable to release scent during immersive experience

2.2 User Interaction

We implemented our VR app in Oculus Quest 2 with native hand gesture control. Our application was written with the Unity3D game engine. We make sure users have a natural hand interaction ability with objects in VR scenes to achieve the most natural interaction, e.g. grab, throw, pick up, stretch. (Fig. 2) We also include realistic visuals, corresponding spatial sound effects and background music

to enhance their sensory experiences besides smell. Particles system animation and renderings were made with Blender3D.



Figure 2: The user is holding an apple in the VR environment, while apple scent is released through the piezoelectric device on the collar.

3 THE IMMERSIVE EXPERIENCE

Many patients with olfactory disorders have not recovered from the disease and are suffering from various inconveniences and psychological challenges brought by it. Among the five senses, the sense of smell may be inconspicuous, but it affects all aspects of life. Smell disorders are invisible disabilities, but they can have a deep impact on people's lives. This short film uses four scenarios to simulate how much change and challenge olfactory disorder will bring to an individual. Integrated with realistic design of 3D models, animation and particle systems, our multi-sensory experience brings the mysterious and intriguing experience of smell disorders to the participants. (Fig. 3a)

The VR experience begins with the first scenario at a family dinner. Participants will start to feel the subtle changes happening in the smell of their food. (Fig. 3b) This will be the introduction for them to notice a difference in their smell ability.

Then, the participants will interact with multiple objects with scents in the house. At this stage, participants will start to experience scent mismatching and distortion, also known as dysosmia. For instance, an orange will start to smell like a mixture with apple scent.

According to patients' feedback, scent mismatching could be confusing and disturbing. They will feel alien to or even disgusted by their favorite scents, including perfume or the smell of their loved ones. So we create the third scenario where the participant will sit next to their partner. Conflicts will be emerged between the patients and their beloved ones. It highlights the needs of communication and psychological care for olfactory dysfunctional patients.

Eventually, as the symptom evolves over time, the participant will lose their sense of smell completely. At this stage, we want to address the hidden danger of smell dysfunction. Therefore, the participants will be facing an emergency where the burnt food in pan causes a fire in the kitchen. Participants need to response to it when pungent scent is no longer available as a hint. (Fig. 3c,d)

4 CONCLUSION

"Living with Smell Dysfunction" is a first of its kind multi-sensory film that mimics smell disorder in VR. By shedding a light on people with invisible diseases, we hope more attention, discussion, research

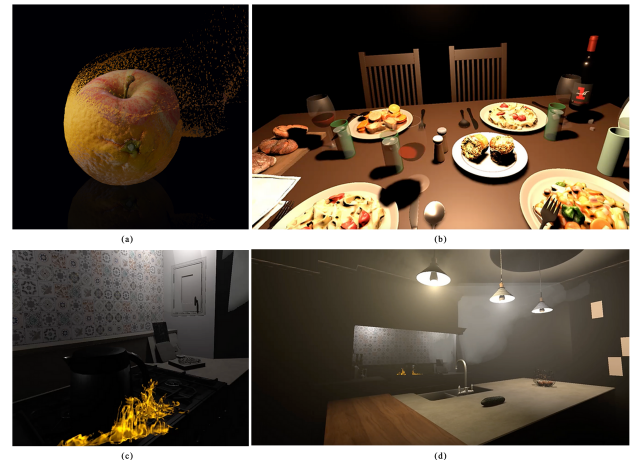


Figure 3: In experience capture: Multi-sensory experience as a smell disorder patient

and treatment could be brought to them. With this, patients could freely share their needs and stories to others.

Our current hardware and its platform could also be utilized in medical fields including smell training, mental therapy and pediatrics treatments in the future. We are connecting research labs targeting Otorhinolaryngology for further research on VR-assisted smell training treatments.

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